

OROVILLE, CA, USA

WMO: 745048

Lat: 39.490N

Lon: 121.618W

Elev: 58

StdP: 100.63

Time Zone: -8.00 (NAP)

Period: 99-19

WBAN: 93210

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	0.6	2.0	-7.7	2.0	10.9	-5.5	2.4	10.0	12.4	10.9	11.2	10.6	1.3	320	0.440

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	16.3	38.0	21.0	36.7	20.7	35.1	20.0	23.0	35.0	22.0	34.1	21.2	33.0	2.8	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
18.6	13.6	30.1	17.3	12.5	29.2	16.3	11.6	27.9	68.2	35.1	64.4	34.1	61.2	33.1	31.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.2	8.1	7.2	-2.2	41.3	1.4	1.3	-3.2	42.3	-4.0	43.1	-4.8	43.8	-5.8	44.8		
(5)			-3.2	25.1	1.2	2.6	-4.0	27.0	-4.7	28.6	-5.4	30.0	-6.3	32.0		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	17.3	8.7	10.6	13.2	15.6	20.0	24.3	27.0	25.7	23.2	18.1	12.2	8.7		
	DBStd	7.12	2.55	2.64	3.04	3.64	3.76	3.68	2.90	2.46	3.10	3.11	3.17	2.76		
	HDD10.0	157	54	22	7	2	0	0	0	0	0	0	15	58		
	HDD18.3	1327	297	215	162	97	24	3	0	0	3	42	184	299		
	CDD10.0	2828	15	40	104	171	310	429	526	488	395	251	82	18		
	CDD18.3	956	0	0	2	16	76	182	267	229	148	34	1	0		
	CDH23.3	12289	1	1	43	255	984	2357	3585	2839	1769	444	11	1		
(13)	CDH26.7	6312	0	0	4	76	404	1247	2080	1521	848	131	1	0		
(14)	Wind	WSAvg	2.7	2.4	3.0	3.2	3.2	3.0	2.7	2.5	2.4	2.3	2.2	2.6		
(15) Precipitation	PrecAvg	530	108	101	76	40	33	9	0	2	2	28	49	114		
	PrecMax	902	358	292	263	124	102	39	2	16	16	94	128	282		
	PrecMin	311	1	0	1	0	1	0	0	0	0	0	0	15		
	PrecStd	151	86	71	81	35	28	11	1	4	4	24	36	69		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	20.1	22.3	27.2	31.8	36.0	39.7	41.0	39.0	37.4	32.8	24.8	19.0		
		MCWB	10.3	13.7	15.8	17.4	20.7	21.7	22.0	20.7	19.8	17.8	14.7	12.1		
	2%	DB	16.6	19.7	24.3	28.8	33.3	37.3	38.6	37.1	35.2	30.0	22.4	16.3		
		MCWB	10.4	12.9	14.7	16.4	19.3	21.0	21.4	20.4	19.0	16.4	14.0	11.7		
	5%	DB	14.7	17.6	22.3	26.5	31.2	35.1	37.1	35.3	33.6	27.7	20.0	14.8		
		MCWB	10.6	11.8	14.0	15.7	18.2	19.8	21.0	19.8	18.7	15.6	13.1	11.2		
	10%	DB	13.1	15.9	19.9	23.7	28.7	33.0	35.3	33.7	31.8	26.0	17.7	13.4		
		MCWB	10.1	11.2	12.7	14.5	17.1	19.1	20.5	19.6	18.4	14.9	12.4	10.4		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	13.7	15.4	16.9	19.2	24.1	24.2	24.5	23.6	22.7	19.4	16.5	14.9		
		MCDB	15.9	19.9	24.6	28.3	31.2	35.8	37.5	33.9	32.0	29.1	22.5	17.4		
	2%	WB	12.3	14.0	15.7	17.5	20.8	22.3	23.0	21.9	20.6	17.9	15.2	13.6		
		MCDB	14.3	17.9	22.7	27.1	30.9	34.3	35.9	33.8	32.3	27.1	19.8	15.3		
	5%	WB	11.5	12.9	14.7	16.2	19.1	21.2	22.1	21.1	19.6	16.9	14.3	12.2		
		MCDB	13.3	16.2	20.6	25.0	29.5	33.0	34.8	32.8	31.3	25.2	17.9	13.5		
	10%	WB	10.6	11.9	13.6	15.1	17.8	20.1	21.3	20.3	18.7	15.8	13.4	11.2		
		MCDB	12.8	14.8	18.7	22.6	27.4	31.5	33.7	31.9	30.5	23.3	16.8	12.6		
(35) Mean Daily Temperature Range	5% DB	MDBR	8.3	9.4	10.9	12.4	13.8	15.2	16.3	16.0	15.4	13.6	10.0	8.2		
		MCDBR	11.7	13.0	15.3	17.0	17.1	17.4	17.4	17.1	17.8	17.0	13.7	10.3		
		MCWBR	7.0	7.6	7.8	7.4	6.9	6.3	6.3	6.3	6.9	7.1	7.0	6.2		
	5% WB	MCDBR	7.8	10.4	13.1	15.8	16.0	16.5	16.5	16.1	16.4	14.5	10.5	6.5		
		MCWBR	5.3	6.6	7.1	7.2	7.7	6.5	6.7	6.6	6.9	6.5	5.8	4.6		
(40) Clear-Sky Solar Irradiance	taub		0.328	0.329	0.351	0.360	0.360	0.356	0.353	0.340	0.343	0.339	0.323	0.327		
	taud		2.485	2.494	2.426	2.397	2.435	2.448	2.439	2.470	2.473	2.486	2.517	2.489		
	Ebn at Noon		841	894	905	915	918	918	917	924	902	869	840	812		
	Edh at Noon		79	89	105	115	113	112	112	106	99	88	75	73		
(44) All-Sky Solar Radiation	RadAvg		2.12	3.05	4.35	5.87	7.25	8.21	8.27	7.36	5.90	4.06	2.52	1.83		
	RadStd		0.38	0.43	0.54	0.58	0.49	0.38	0.26	0.25	0.23	0.32	0.23	0.33		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (32 neighbors)	+0.41	N/A	-1.11	N/A	N/A	N/A	-32	-100	+134	+50

Nomenclature: See separate page